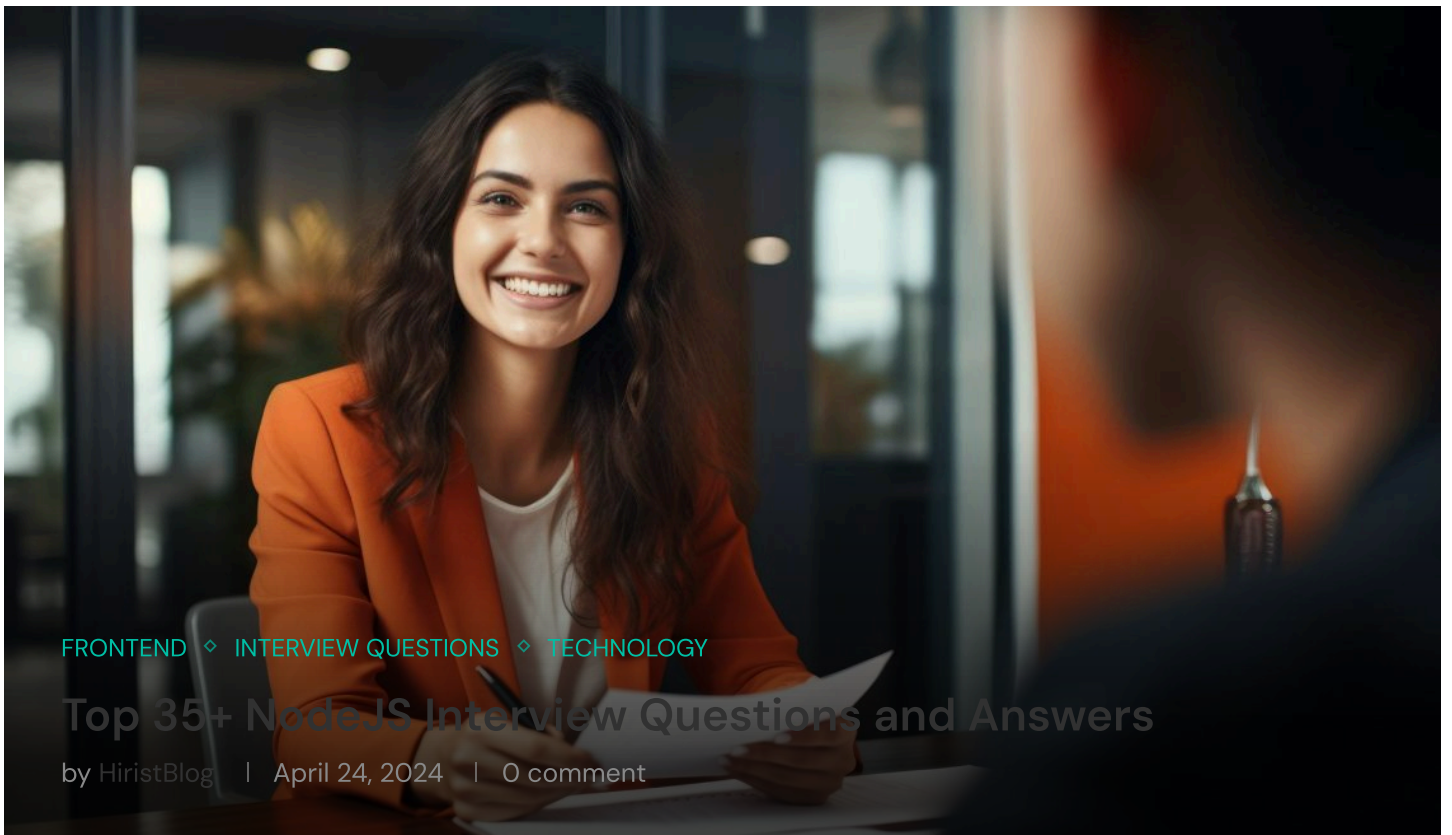




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Top 35+ NodeJS Interview Questions and Answers

by HirstBlog | April 24, 2024 | 0 comment

Are you preparing for a Node.js interview and wondering what questions might come your way? Well, we can help you with this! In this article, we've compiled the top 25+ **NodeJS interview questions** along with their answers to help you ace your interview with confidence. Whether you're new to Node.js or experienced, these questions cover important topics like asynchronous programming, modules, event loops, callbacks, and more.

So, let's begin with the **Node JS interview question and answer!**

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A Little About Node JS

Node.js is a runtime environment that lets you run JavaScript code outside of a web browser.

It's built on the V8 JavaScript engine and uses an event-driven, non-blocking I/O model, making it efficient for building scalable and high-performance applications.

Node.js is widely used for web development to create server-side applications and APIs.

Here are some stats highlighting the popularity of Node JS:

- It is used by over 30 million websites.
- More than 262,196 companies use Node.js.
- Companies like PayPal, Netflix, LinkedIn, Amazon, Reddit, eBay, and others use Node.js.
- 42.73% of professional developers prefer Node.js.

In India, there's a high demand for Node.js developers. Here are some top IT companies offering attract packages for this position:

- [Tata Consultancy Services](#)
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Now, if you are looking for the latest [Node JS job openings](#), visit HIRIST. It is an online IT job portal where you can easily find the top IT jobs in India.

Node JS Interview Questions for Fresher

Here are some important **Node JS basic interview questions** and answers:



1. What is npm?

npm (Node Package Manager) is a package manager for Node.js that allows you to install, manage, and share reusable JavaScript code packages or modules.

2. How do you install Node.js on your computer?

You can download and install Node.js from the official Node.js website. Choose the appropriate installer for your operating system and follow the installation steps.

3. What is a module in Node.js?

A module in Node.js is a JavaScript file that contains reusable code. Modules help in organizing and maintaining code by encapsulating related functionality.

4. How do you import modules in Node.js?

You can import modules in Node.js using the `require()` function.

For example:

```
const fs = require('fs');
```

5. What is an event-driven programming model in Node.js?

Node.js uses an event-driven, non-blocking I/O model. This means that Node.js is designed to handle multiple operations concurrently without waiting for one operation to finish before starting another.

6. What is the most popular Node.js framework?

The most popular Node.js framework is [Express.js](#). It is widely used for building web applications and APIs due to its simplicity, flexibility, and robust features.

Also Read – [Top 25+ JavaScript Interview Questions and Answers](#)

Node JS Interview Questions for Experienced Professionals

Here are some commonly asked **interview questions and answers on Node JS** for experienced candidates:



NodeJS Interview Questions for 2 Years Experience

7. What is middleware in Express.js?

Middleware in Express.js are functions that have access to the request and response objects. They can modify these objects, execute additional code, or terminate the request-response cycle.

8. Explain the concept of streams in Node.js.

Streams in Node.js are objects used to read or write data sequentially. They allow you to process large amounts of data in chunks, which is memory-efficient and suitable for handling data from files, network connections, or other sources.

9. How do you handle CORS (Cross-Origin Resource Sharing) in an Express.js application?

CORS can be handled using the cors middleware in Express.js, which allows or restricts cross-origin HTTP requests based on specified configurations.

10. Explain the difference between callbacks and Promises in Node.js.

Callbacks are functions passed as arguments to other functions and executed asynchronously. Promises are objects representing the eventual completion or failure of an asynchronous operation.

NodeJS Interview Questions for 3 Years Experience

11. What is clustering in Node.js?

Clustering in Node.js refers to the technique of spawning multiple child processes (workers) to take advantage of multi-core systems. It improves performance and scalability by distributing the workload across multiple CPU cores.

12. How does Node.js handle asynchronous operations?

Node.js uses event-driven, non-blocking I/O to handle asynchronous operations efficiently. It uses event loops and callbacks to manage multiple concurrent operations without blocking the execution thread.

13. What is the Event Emitters concept in Node.js, and how do you use it?

Event Emitters in Node.js facilitate communication between objects using an event-driven architecture. You can create custom event emitters using the events module to emit and listen to events.

14. Explain how you would implement authentication and authorization in a Node.js application.

Authentication can be implemented using middleware like [Passport.js](#) and JWT (JSON Web Tokens). Authorization involves defining roles and permissions to restrict access to certain resources.

NodeJS Interview Questions for 5+ Years' Experience

15. How does the Node.js event loop work, and what are its phases?

The Node.js event loop is responsible for handling asynchronous operations. It consists of multiple phases (timers, pending callbacks, idle/prepare, poll, check, and close callbacks) that manage the execution of callbacks.

16. Discuss the pros and cons of using Node.js for real-time applications like chat applications and gaming servers.

Node.js excels in real-time applications due to its event-driven, non-blocking architecture, making it suitable for handling multiple concurrent connections. However, it may not be ideal for CPU-intensive tasks.

17. Explain how you would optimize a Node.js application for performance and scalability.

Performance optimization techniques include using caching mechanisms (e.g., Redis), implementing load balancing, profiling and optimizing critical code paths, and leveraging asynchronous patterns to handle concurrency effectively.

18. How do you handle errors in Node.js applications?

This is one of the most important **Node JS problem solving questions**. Here's how you should answer

Errors in Node.js can be handled using try-catch blocks for synchronous code and `.catch()` method for Promises. You can also use middleware functions with Express.js to handle errors in web applications.

See also [Top 25 iOS Interview Questions and Answers](#)

Advanced Node JS Interview Questions

Learn these advanced **NodeJS interview questions** and answers:

19. How does garbage collection work in Node.js?

Node.js uses the V8 engine's garbage collector to manage memory. It automatically detects and removes unused objects to free up memory.

20. Discuss the use cases and benefits of using `async_hooks` in Node.js.



`async_hooks` is a module that allows you to monitor asynchronous resource usage in Node.js, useful for debugging, tracing, and profiling applications.

21. How would you implement caching in a Node.js application?

Caching can be implemented using in-memory caches like Redis or Memcached to store frequently accessed data, reducing the need for expensive database or API calls.

22. What is the purpose of the `util.promisify()` function in Node.js?

`util.promisify()` is a utility function used to convert callback-based asynchronous functions into Promises, allowing easier integration with `async/await` syntax.

23. Discuss the use of worker threads in Node.js for parallel processing.

Worker threads in Node.js enable parallel execution of JavaScript code by creating separate threads that can perform CPU-intensive tasks concurrently, leveraging multi-core CPUs for improved performance.

Node JS Coding Interview Questions

Here are some coding-related **Node JS developer interview questions** and answers:

24. Write a function to reverse a string in Node.js.

```
function reverseString(str) {  
  
    return str.split("").reverse().join("");  
  
}
```

// Example usage:

```
console.log(reverseString('hello')); // Output: 'olleh'
```

25. Implement a function to check if a number is prime in Node.js.

```
function isPrime(num) {  
  
    if (num <= 1) return false;  
  
    if (num <= 3) return true;  
  
    if (num % 2 === 0 || num % 3 === 0) return false;
```



```
for (let i = 5; i * i <= num; i += 6) {  
  
    if (num % i === 0 || num % (i + 2) === 0) {  
  
        return false;  
  
    }  
  
}  
  
return true;  
  
}
```

// Example usage:

```
console.log(isPrime(11)); // Output: true
```

26. Create a function to calculate the sum of all elements in an array in Node.js.

```
function sumArray(arr) {  
  
    return arr.reduce((total, current) => total + current, 0);  
  
}
```

// Example usage:

```
const numbers = [1, 2, 3, 4, 5];  
  
console.log(sumArray(numbers)); // Output: 15
```

27. Write a function to sort an array of numbers in ascending order in Node.js.

```
function sortArray(arr) {  
  
    return arr.sort((a, b) => a - b);  
  
}
```

// Example usage:

```
const unsortedArray = [3, 1, 5, 2, 4];  
  
console.log(sortArray(unsortedArray)); // Output: [1, 2, 3, 4, 5]
```

Node JS Backend Interview Questions



28. What is the purpose of Express.js in Node.js backend development?

Express.js is a minimalist and flexible Node.js web application framework that provides a robust set of features for building web servers and APIs. It simplifies routing, middleware integration, and request handling.

29. Explain the difference between `app.get()` and `app.post()` in Express.js routing.

- `app.get()` is used to handle HTTP GET requests.
- `app.post()` is used to handle HTTP POST requests. Both methods define routes and specify callback functions to handle incoming requests for the specified route.

30. How do you handle POST requests in Express.js to receive and process form data?

You can use the `body-parser` middleware to parse incoming request bodies.

Here's an example of handling a POST request:

```
const express = require('express');
```

```
const bodyParser = require('body-parser');
```

```
const app = express();
```

See also [Top 30 Django Interview Questions Answers \(2024\)](#)

```
app.use(bodyParser.urlencoded({ extended: true }));
```

```
app.post('/submit', (req, res) => {
```

```
    const formData = req.body;
```

```
    // Process form data here
```

```
    res.send('Form data received');
```

```
});
```

```
app.listen(3000, () => {
```

```
    console.log('Server is running on port 3000');
```



```
});
```



Node JS Scenario Based Questions

Here are some important **Node JS scenario based questions** and answers:

31. You have a Node.js application that needs to read data from a JSON file named data.json and log it to the console. How would you achieve this?

To read data from data.json and log it to the console in a Node.js application, you can use the fs (file system) module which is built into Node.js.

First, require the fs module at the beginning of your script. Then, use fs.readFile to asynchronously read the file content, and inside the callback function, parse the JSON data and log it to the console.

```
const fs = require('fs');

fs.readFile('data.json', 'utf8', (err, data) => {

  if (err) {

    console.error(err);

    return;

  }

  const jsonData = JSON.parse(data);

  console.log(jsonData);

});
```

32. Suppose you want to create a simple HTTP server using Node.js. How would you implement this?

To create a basic HTTP server in Node.js, you can use the http module. First, require the http module and then use the createServer method to create a server instance. Inside the server's request listener, you can handle incoming HTTP requests and send back responses.

```
const http = require('http');

const server = http.createServer((req, res) => {

  res.writeHead(200, { 'Content-Type': 'text/plain' });
```

```
res.end('Hello World!');

});

const PORT = 3000;

server.listen(PORT, () => {

  console.log(`Server is running on port ${PORT}`);

});
```



TCS Node JS Interview Questions

Here are some commonly asked TCS **NodeJS interview questions** and answers:

33. How would you contribute to TCS as a Node.js developer?

"As a Node.js developer, I'll use my skills to create strong and scalable applications that meet TCS's client needs. I'll focus on writing good code, following best practices, and always learning new technologies to bring innovation to the team."

34. Can you describe a challenging project you've worked on related to Node.js?

"In one of my past projects, I built a live chat app using Node.js, [Socket.io](#), and [MongoDB](#). I faced challenges setting up message syncing, user login, and data storage, but I solved them by planning carefully and working closely with my team."

Infosys Node JS Interview

Here are some important Infosys **Node JS questions** and answers:

35. Why are you interested in joining Infosys as a Node.js developer?

"Infosys is a well-known global IT company that focuses on innovation and serving clients well. I'm thrilled about the chance to use my Node.js skills at Infosys, work with diverse teams, and contribute to important projects that push technology forward."

36. Can you share a project where you utilized Node.js effectively to solve a business challenge?

"In a past project, I created a web app using Node.js and Express.js to help a retail client manage their inventory more efficiently."

By using Node.js's features that handle multiple tasks at once and breaking the project into smaller parts, I made it easier to get data quickly and update it in real-time. This led to better control of

inventory and improved how the business operated.”



Also Read – [Top 25+ Frontend Interview Questions and Answers](#)

Wrapping Up

So, these are the top 25+ **NodeJS interview questions** and answers to help you [prepare for your next interview](#). Understanding these concepts will help you answer all the questions with confidence. And now if you are searching for Node.js job opportunities, visit [Hirist](#). It is an IT job portal where you can easily find exciting Node.js jobs and boost your career in this dynamic field.

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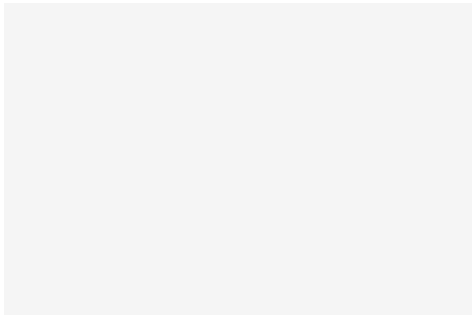
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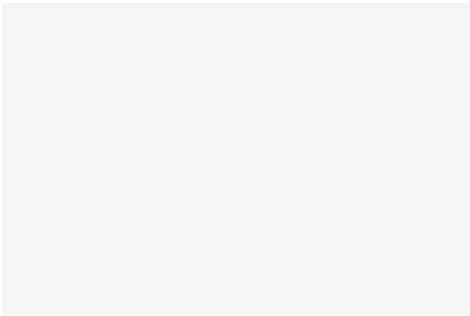
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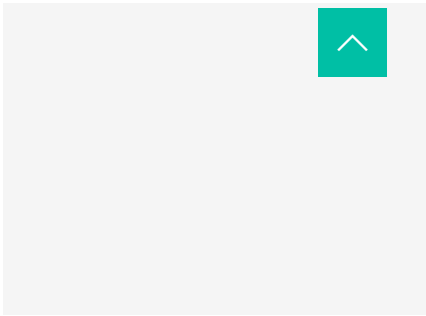
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